Abhronil Paul

Electronics Design Engineer — Avionics & Power Subsystems

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Summary

Electronics hardware engineer with three space-sector roles designing and validating PCBs from schematic to bring-up, including a sensor-interface board built at Capsule Corporation that is flying on an upcoming In-Orbit Demonstration mission. Comfortable owning a board end-to-end: Altium/KiCad layout, bench characterization, and failure analysis (FMEA/FTA) that catches design margins before they become field failures.

Education

Politecnico di Milano — **2023 – 2026**
MSc Electronics Engineering — *Milano, Italy*

Thapar Institute of Engineering & Technology — **2018 – 2022**
BEng Electronics & Communication Engineering | Academic Merit Scholarship — *Patiala, India*

Technical Skills

EDA & Simulation: Altium Designer, KiCad, LTspice, MATLAB/Simulink, Xilinx Vivado.
Interfaces & Protocols: SPI, I2C, UART, CAN, RS-232, RS-485.
Risk Analysis: FMEA, FMECA, FTA, failure propagation analysis, criticality assessment, single-point-of-failure identification.
Software & Programming: Python, C, Bash, Git, test automation, parametric trending, tool development.
Standards & Compliance: ECSS-E-ST-10-02, ECSS-Q-ST-60, NASA-STD-8729.1A, MIL-HDBK-217, IEC 61508, IEEE 1233.

Professional Experience

Lunar Outpost EU — **Sep. 2025 – Feb. 2026**
Electronics Design Engineer — *Luxembourg*

- Delivered a flight-candidate 75 V power distribution unit by designing an 8-layer schematic and PCB layout, selecting MOSFETs, inductors, and capacitors for thermal and switching margins, and closing signal/power integrity issues found in layout review.
- Validated PDU performance under thermal stress by building a bench test setup with an oscilloscope and thermal chamber, characterizing circuit behavior across the operating range, and iterating the design to close two margin failures before sign-off.
- Kept the design auditable through ECSS-aligned documentation by maintaining requirement traceability and compliance matrices across every design revision.

Revolv Space S.r.l. — **Jan. 2025 – Jul. 2025**
Electronics Systems Engineer — *Torino, Italy*

- Qualified the TI DRV8889-Q1 motor driver for flight use by designing custom test circuits, developing parametric characterization protocols, and running bench measurements against datasheet limits.
- Surfaced three latent design margins by authoring a component-level FMEA on the power stage and cross-checking failure modes against measured stress data.
- Closed the hardware-firmware interface gap by coordinating with the embedded team on signal timing and electrical compatibility, debugging mismatches with a logic analyzer.

Capsule Corporation S.r.l. — **Jan. 2024 – Jan. 2025**
Electronics Integration Engineer — *Milano, Italy*

- Built a sensor-interface PCB now flying on an upcoming In-Orbit Demonstration mission by designing analog signal-conditioning schematics, selecting precision amplifiers and ADC components, and optimizing layout for noise-sensitive traces.
- Resolved multi-layer design defects that blocked integration by isolating root cause on the bench with an oscilloscope and multimeter, then iterating the schematic through two revisions.
- Kept the board buildable across three design revisions by managing component lifecycle, tracking obsolescence, and qualifying alternate parts.

STMicroelectronics Pvt. Ltd. — **Jan. 2022 – Jul. 2022**
Electronics R&D Engineer — *Greater Noida, India*

- Characterized power semiconductor switching and thermal limits by designing test circuits and running them against SPICE simulation, producing data that informed component selection for downstream system design.
- Shortened the design-review cycle by packaging switching-loss and thermal-limit findings into technical reports used directly by the system design team.

Projects

Fault-Tolerant Supervisor Architecture for Motor Drivers | *KiCAD, FMEA, LTspice* — **2025**

- MSc thesis: architected 5-channel analog supervisor system (block-level design, failure propagation analysis, FTA); formulated watchdog and fault-latch requirements; validated architecture via fault-injection testing.

FPGA-Based CNN Accelerator for Debris Detection | *Xilinx Vivado, Verilog, Python* — **2024**

- Architected real-time on-board processing subsystem; designed data-flow diagrams and interface specifications; implemented resource-constrained hardware architecture.

Recognition

- **Space Cadet, EnduroSat** — selected into a global CubeSat engineering cohort; owned systems-level design under resource constraints alongside engineers from 20+ countries.
- **ESA FYS Booster Programme** — one of a small cohort advancing a CubeSat concept through ESA's technical review process.
- **Takeoff Accelerator 2024** — sole electronics lead on a hardware startup team; owned requirements, architecture, and risk analysis under investor deadline pressure.